IILM UNIVERSITY, GREATER NOIDA
SCHOOL OF COMPUTER SCIENCE & ENGINEERING
WEEKLY PROGRESS REPORT
Even semester 2026-2027

B.Tech. Major Project Details

Project Id	26P_3CSE024
Project Title	AI-Based Precision Agriculture and Crop Health Monitoring System
Reporting period	14/10/2026 to 20/10/2026
Week No.	3

Work Completed This Week (Phase III: Implementation, Part 1):

- **Prototype / Initial Implementation:** Started implementation with two AI modules: a leaf-image plant disease classifier (completed) and a first prototype of the crop yield predictor for Haryana districts.
- **Data Preparation:** Prepared the PlantVillage dataset (54,305 images), checked for duplicate images, and split it into train, validation and test sets before augmentation to avoid data leakage.
- **Design & Implementation:** Built the disease model using transfer learning (MobileNetV2, 38 classes) in TensorFlow/Keras, trained in two stages (new classifier head, then fine-tuning), chosen for its light weight and limited GPU access.
- **Design Validation:** Training and validation accuracy stayed close (98.3% vs 97.5%), showing no serious over-fitting; the best checkpoint was selected on validation data only.
- **Experimental Results:** Test accuracy of 97.6% (macro F1 0.97) on unseen images; weakest class was Tomato Early blight. Low-confidence predictions are flagged for the user.
- **Coding Quality & Documentation:** Wrote a reusable inference script that returns JSON output, with saved configuration, class list, requirements and README, so the dashboard can call the model directly.
- **Yield Prototype:** Prepared district-wise seasonal data for Wheat, Mustard and Rice (2010–2022) and compared a baseline with Random Forest and XGBoost; weather-only models did not beat the simple baseline, so the approach is to be improved.

Challenges/Issues:

- Difficulties faced: PlantVillage images are lab-style, so accuracy on real field photos is not yet verified; yield data is small and weather alone gives weak predictions.
- Action taken/proposed: test set used only once, confidence warning added, and historical yield features to be added to the yield model.

Plan for Next Week:

- Improve the yield model using previous-year yield data.
- Design and build the irrigation advisory module.
- Test all modules and prepare a common input/output format for integration.
- Discuss progress and the mid-term evaluation process with the Project Guide.

Guide's Remarks:

Signature of Student

Signature of Guide

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B.Tech. Major Project Details

Project Id	26P_3CSE024
Project Title	AI-Based Precision Agriculture and Crop Health Monitoring System
Reporting period	21/10/2026 to 27/10/2026
Week No.	4

Work Completed This Week (Phase III: Implementation, Part 2):

- **Yield Model Improvement:** Added previous-year yield features to the Random Forest model; it now outperforms the baselines (R^2 0.92), working best for Rice and less well for Wheat and Mustard.
- **Irrigation Module – Design & Implementation:** Built an irrigation advisor for Wheat, Mustard and Rice. As no public soil-moisture dataset exists for Haryana, soil moisture was simulated with a standard FAO-56 water-balance method using real weather data; a Random Forest decides "irrigate today or not" and a rule gives the water amount (Rice uses a standing-water rule).
- **Testing & Validation:** Tested on unseen years and unseen districts with sensor and rain-forecast noise added; confirmed the inference scripts for the yield and irrigation modules run correctly.
- **Performance Analysis:** The irrigation advisor saved about 21% water for Mustard against a fixed schedule and reduced crop-stress days for Wheat, though its advantage over a tuned sensor rule is modest.
- **System Integration:** The disease, yield and irrigation modules are each packaged as independent components with inference scripts, ready to be connected through the recommendation layer and dashboard.
- **Discussion with Project Guide:** Presented the progress of the three modules and discussed the submission of the Google Form for the mid-term evaluation and the implementation to be shown.
- **Milestone Completion:** AI model development milestone achieved for disease detection, yield prediction and irrigation advisory; recommendation layer and dashboard are pending.

Challenges/Issues:

- Difficulties faced: irrigation results are based on simulated soil moisture and not yet validated with real sensor data.
- Action taken/proposed: limitations documented; real sensor readings to be logged and the model retrained when available.

Plan for Next Week:

- Build the recommendation layer combining disease, irrigation and yield outputs.
- Develop the dashboard (backend and web interface) with alerts.
- Perform integration testing and submit the Google Form for the mid-term evaluation.

Guide's Remarks:

Signature of Student

Signature of Guide